CSE412 Project Report

Independent Work Distribution

Team Members

1. Arunavo
2. Talha
3. Mahir
4. Walid

ARUNAVO

Assigned Report Sections

1. Group Information

2. Project Name

3. Client Information

4. Client Working Experience with the Team

5. Client Satisfaction

6. Team Experience Working with the Client and Project

7. Requirements Analysis

8. Functional Requirements

9. Non Functional Requirements

10. Acceptance Criteria

11. Key Features of the Application

12. Overall Architecture

13. Architecture Diagram

14. Sequence Diagram

15. Architecture Pattern

16. Design Patterns of the System

17. Detailed Sprint Work

Arunavo documents all sprint work independently from the repository, project history, and available development records.

19. Final Application Development Summary

Explain how the application was developed from requirements through implementation.

Arunavo's Diagrams

1. Architecture Diagram
2. Sequence Diagram

Arunavo's Boundary

Arunavo does not perform:

- Function Point calculations
- Source Lines of Code calculations
- Thousand Source Lines of Code calculations
- Constructive Cost Model calculations
- Cost driver calculations
- Effort Adjustment Factor calculations
- Detailed Constructive Cost Model calculations
- Rayleigh calculations
- Project scheduling
- Gantt Chart
- Testing documentation

Arunavo's work is completely based on the project and repository and does not require results from another member.

TALHA

Assigned Report Sections

1. Data Flow Diagram Level 0,1 & 2

2. State Machine Diagram

3. Use Case Diagram

4. Unified Modeling Language Diagram

5. Database Entity Relationship Diagram

6. Simple Application Flow Diagram

7. Overall Application Explanation

Explain the complete application independently by examining the repository.

8. Application Functionality by User Role

Document every relevant user role and its available functionality.

9. Page Navigation by User Role

Document the actual navigation structure for every user role.

10. Final Application Workflow

Document the complete application workflow.

11. Project Scheduling

Develop the project scheduling section independently.

12. Product Breakdown Structure or Hybrid Work Breakdown Structure

Prepare the complete project work breakdown.

13. Gantt Chart

Prepare the project Gantt Chart based on the actual project activities and ten-week project duration.

14. Total Time Spent Each Week by the Team

Document the team's weekly time information using available project records.

Talha's Diagrams

1. Data Flow Diagram Level 0,1 & 2
2. State Machine Diagram
3. Use Case Diagram
4. Unified Modeling Language Diagram
5. Database Entity Relationship Diagram
6. Simple Application Flow Diagram
7. Product Breakdown Structure or Hybrid Work Breakdown Structure
8. Gantt Chart

Talha's Boundary

Talha does not perform:

- Function Point calculations
- Source Lines of Code calculations
- Thousand Source Lines of Code calculations

- Constructive Cost Model calculations
- Cost driver calculations
- Effort Adjustment Factor calculations
- Detailed Constructive Cost Model calculations
- Rayleigh calculations
- Testing documentation
- Client documentation
- Requirements documentation
- Sprint documentation

Talha's work is independently produced from the repository and project records.

MAHIR

Assigned Report Sections

Mahir owns the complete **software measurement and Function Point estimation section**.

1. Software Measurement Overview (Chapter 8)

2. Direct and Indirect Measures (Chapter 8)

3. Size Oriented Metrics (Chapter 8)

4. Lines of Source Code and Thousand Lines of Source Code (Chapter 8)

5. Function Oriented Metrics (Chapter 8)

6. Function Point Analysis (Chapter 8)

7. External Inputs, External Outputs, External Inquiries, Internal Logical Files, and External Interface Files (Chapter 8)

8. Unadjusted Function Point Calculation (Chapter 8)

9. Complexity Adjustment Factors (Chapter 8)

10. Adjusted Function Point Calculation (Chapter 8)

11. TypeScript Language Factor Assumption and Justification (Chapter 8)

12. Source Lines of Code Calculation (Chapter 8)

13. Thousand Source Lines of Code Calculation (Chapter 8)

Mahir's Chapter 8 Responsibility

Mahir independently performs the complete Function Point and software-size analysis by inspecting the repository.

He produces:

- Unadjusted Function Point
- Complexity Adjustment Value
- Adjusted Function Point
- TypeScript Language Factor
- Source Lines of Code
- Thousand Source Lines of Code

No other member needs to calculate these metrics.

Mahir's Diagrams

1. None of the mandatory system diagrams are assigned to Mahir because his complete Function Point and software-size calculation is his primary analytical workload.

Mahir's Boundary

Mahir does not perform:

- Constructive Cost Model category selection
- Basic Constructive Cost Model
- Staff Size
- Productivity
- Cost Drivers
- Effort Adjustment Factor
- Intermediate Constructive Cost Model
- Detailed Constructive Cost Model
- Rayleigh Model
- Testing
- Scheduling
- Gantt Chart
- Requirements documentation
- Client documentation
- Sprint documentation
- Application walkthrough

Mahir's entire assignment is independently completed from the repository and Chapter 8.

WALID

Assigned Report Sections

Walid owns the complete **Constructive Cost Model and testing section**.

1. Testing Overview

2. Total Test Cases

3. Types of Test Cases

4. Primary and Major Test Cases

5. Constructive Cost Model Category Selection (Chapter 8)

6. Basic Constructive Cost Model (Chapter 8)

7. Basic Constructive Cost Model Effort Calculation (Chapter 8)

8. Basic Constructive Cost Model Development Time Calculation (Chapter 8)

9. Staff Size Calculation (Chapter 8)

10. Productivity Calculation (Chapter 8)

11. Cost Drivers (Chapter 8)

12. Product Attributes (Chapter 8)

13. Computer and Hardware Attributes (Chapter 8)

14. Personnel Attributes (Chapter 8)

15. Project Attributes (Chapter 8)

16. Cost Driver Ratings and Multipliers (Chapter 8)

17. Effort Adjustment Factor (Chapter 8)

18. Intermediate Constructive Cost Model (Chapter 8)

19. Intermediate Constructive Cost Model Effort Calculation (Chapter 8)

20. Intermediate Constructive Cost Model Development Time Calculation (Chapter 8)

21. Detailed Constructive Cost Model (Chapter 8)

22. Principle of Effort Estimation (Chapter 8)

23. Lifecycle Effort Distribution (Chapter 8)

24. Lifecycle Schedule Distribution (Chapter 8)

25. Size Equivalent Calculation, if Applicable (Chapter 8)

26. Phase Wise Effort Distribution (Chapter 8)

27. Phase Wise Development Time Distribution in Weeks (Chapter 8)

28. Ten Week Project Schedule Comparison (Chapter 8)

29. Rayleigh Model (Chapter 8)

30. Rayleigh Effort Rate and Staffing Calculation (Chapter 8)

31. Rayleigh Cumulative Effort Calculation (Chapter 8)

32. Rayleigh Peak Staffing Calculation (Chapter 8)

Walid's Diagrams and Tables

1. Testing Workflow Diagram
2. Cost Estimation Flow Diagram
3. Cost Driver and Effort Adjustment Factor Table

Walid's Boundary

Walid independently performs the complete cost estimation chain.

He does not depend on Mahir's final numerical result.

He independently analyzes the repository and performs the required software-size reasoning necessary for his Constructive Cost Model calculation.

He does not modify Mahir's Function Point section.

Walid does not perform:

- Client documentation
- Requirements documentation
- Architecture documentation
- UML modelling
- Database Entity Relationship Diagram
- Application walkthrough
- Sprint documentation
- Project scheduling
- Gantt Chart

FINAL REPORT SEQUENCE

The merged report must follow this order:

1. Group Information

2. Project Name

3. Client Information

4. Client Working Experience with the Team

5. Client Satisfaction

6. Team Experience Working with the Client and Project

7. Requirements Analysis

8. Functional Requirements

9. Non Functional Requirements

10. Acceptance Criteria

11. Key Features of the Application

12. Overall Architecture

13. Architecture Diagram

14. Sequence Diagram

15. Data Flow Diagram Level 0

16. Data Flow Diagram Level 2

17. State Machine Diagram

18. Use Case Diagram

19. Unified Modeling Language Diagram

20. Database Entity Relationship Diagram

21. Simple Application Flow Diagram

22. Architecture Pattern

23. Design Patterns of the System

24. Testing Overview

25. Total Test Cases

26. Types of Test Cases

27. Primary and Major Test Cases

28. Detailed Sprint Work

29. Total Time Spent Each Week by the Team

30. Software Measurement Overview (Chapter 8)

31. Direct and Indirect Measures (Chapter 8)

32. Size Oriented Metrics (Chapter 8)

33. Lines of Source Code and Thousand Lines of Source Code (Chapter 8)

34. Function Oriented Metrics (Chapter 8)

35. Function Point Analysis (Chapter 8)

36. External Inputs, External Outputs, External Inquiries, Internal Logical Files, and External Interface Files (Chapter 8)

37. Unadjusted Function Point Calculation (Chapter 8)

38. Complexity Adjustment Factors (Chapter 8)

39. Adjusted Function Point Calculation (Chapter 8)

40. TypeScript Language Factor Assumption and Justification (Chapter 8)

41. Source Lines of Code Calculation (Chapter 8)

42. Thousand Source Lines of Code Calculation (Chapter 8)

43. Constructive Cost Model Category Selection (Chapter 8)

44. Basic Constructive Cost Model (Chapter 8)

45. Basic Constructive Cost Model Effort Calculation (Chapter 8)

46. Basic Constructive Cost Model Development Time Calculation (Chapter 8)

47. Staff Size Calculation (Chapter 8)

48. Productivity Calculation (Chapter 8)

49. Cost Drivers (Chapter 8)

50. Product Attributes (Chapter 8)

51. Computer and Hardware Attributes (Chapter 8)

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60. Principle of Effort Estimation (Chapter 8)

61. Lifecycle Effort Distribution (Chapter 8)

62. Lifecycle Schedule Distribution (Chapter 8)

63. Size Equivalent Calculation, if Applicable (Chapter 8)

64. Phase Wise Effort Distribution (Chapter 8)

65. Phase Wise Development Time Distribution in Weeks (Chapter 8)

66. Ten Week Project Schedule Comparison (Chapter 8)

67. Rayleigh Model (Chapter 8)

68. Rayleigh Effort Rate and Staffing Calculation (Chapter 8)

69. Rayleigh Cumulative Effort Calculation (Chapter 8)

70. Rayleigh Peak Staffing Calculation (Chapter 8)

71. Project Scheduling

72. Product Breakdown Structure or Hybrid Work Breakdown Structure

73. Gantt Chart

74. Overall Application Explanation

75. Application Functionality by User Role

76. Page Navigation by User Role

77. Final Application Workflow

FINAL OWNERSHIP SUMMARY

Member	Main Responsibility
Arunavo	Project Information, Requirements, Architecture, Architecture Pattern, Design Patterns, Sprint Work
Talha	System Modelling, Application Explanation, User Roles, Navigation, Project Scheduling, Work Breakdown Structure, Gantt Chart
Mahir	Complete Software Measurement and Function Point Analysis (Chapter 8)
Walid	Complete Testing and Cost Estimation from Constructive Cost Model through Rayleigh (Chapter 8)

Each member can work independently using the repository.

No member is required to wait for another member to finish their calculations.

The final report is created by placing each member's completed sections into the **Final Report Sequence** above.